

LECTURES 5 - 6

Theorem 0.1 Let $\{X_n\}$ be a sequence of random variables on $(\Omega, \mathcal{F})$ such that

$$\lim_{n \rightarrow \infty} X_n(\omega) \text{ exists for all } \omega \in \Omega.$$

Then $X : \Omega \rightarrow \mathbb{R}$ defined by

$$X(\omega) = \lim_{n \rightarrow \infty} X_n(\omega), \quad \omega \in \Omega$$

is a random variable.

Proof. For $a \in \mathbb{R}$.

$$\begin{aligned} \omega \in \{X \leq a\} &\Rightarrow \text{for each } m \geq 1, \text{ there exists } n \text{ such that} \\ &X_k(\omega) \leq X(\omega) + \frac{1}{m} \leq a + \frac{1}{m} \text{ for all } k \geq n \end{aligned}$$

$$\Rightarrow \omega \in \bigcup_{n=1}^{\infty} \bigcap_{k=n}^{\infty} \left\{ X_k \leq a + \frac{1}{m} \right\}, m \geq 1$$

$$\Rightarrow \omega \in \bigcap_{m=1}^{\infty} \bigcup_{n=1}^{\infty} \bigcap_{k=n}^{\infty} \left\{ X_k \leq a + \frac{1}{m} \right\}.$$

Hence

$$\{X \leq a\} \subseteq \bigcap_{m=1}^{\infty} \bigcup_{n=1}^{\infty} \bigcap_{k=n}^{\infty} \left\{ X_k \leq a + \frac{1}{m} \right\}.$$

Now for

$$\omega \in \bigcap_{m=1}^{\infty} \bigcup_{n=1}^{\infty} \bigcap_{k=n}^{\infty} \left\{ X_k \leq a + \frac{1}{m} \right\}$$

$$\Rightarrow \omega \in \bigcup_{n=1}^{\infty} \bigcap_{k=n}^{\infty} \left\{ X_k \leq a + \frac{1}{m} \right\}, \text{ for all } m \geq 1$$

$$\Rightarrow \text{for each } m \geq 1, X_k(\omega) \leq a + \frac{1}{m} \\ \text{for all } k \geq n_1 \text{ for some } n_1$$

$$\Rightarrow X(\omega) \leq a. \text{ (exercise : use the definition of limit)}$$

Hence we have

$$\{X \leq a\} = \bigcap_{m=1}^{\infty} \bigcup_{n=1}^{\infty} \bigcap_{k=n}^{\infty} \left\{ X_k \leq a + \frac{1}{m} \right\}. \quad (0.1)$$

Since $\{X_k \leq a + \frac{1}{m}\} \in \mathcal{F}$ for all k, m and $\mathcal{F}$ is a σ -field, from (0.1) it follows from the definition of σ -field that

$$\{X \leq a\} \in \mathcal{F}.$$

Therefore X is a random variable.

Definition 0.1 A function $f : \mathbb{R} \rightarrow \mathbb{R}$ is said to be Borel (measurable) function if $f^{-1}(B) \in \mathcal{B}_{\mathbb{R}}$ for all $B \in \mathcal{B}_{\mathbb{R}}$.

Example 0.1 If $f : \mathbb{R} \rightarrow \mathbb{R}$ is continuous, then it is Borel function. This follows from the fact that $f^{-1}(O)$ is open if O is open in $\mathbb{R}$.

Hint: $\mathcal{D} = \{B \in \mathcal{B}_{\mathbb{R}} \mid f^{-1}(B) \in \mathcal{B}_{\mathbb{R}}\}$ is a σ -field and contains all open sets.

Example 0.2 If $f : \mathbb{R} \rightarrow \mathbb{R}$ is a monotone function (i.e. either increasing¹ or decreasing), then f is a Borel function.

Let f be increasing. For $a \in \mathbb{R}$, consider the set $D_a := \{f > a\}$.

Set

$$c = \inf\{x \mid f(x) > a\}.$$

There are two cases: $c \in D_a$ or $c \notin D_a$.

When $c \in D_a$ (then $c > -\infty$).

$$x \geq c \Rightarrow f(x) \geq f(c) > a \Rightarrow x \in D_a.$$

Hence $[c, \infty) \subseteq D_a$. From the definition of D_a and c , it follows that $D_a \subseteq [c, \infty)$. Hence $D_a = [c, \infty) \in \mathcal{B}_{\mathbb{R}}$.

When $c \notin D_a$ (then it is possible that $c = -\infty$: for example $f(x) = e^x, x \in \mathbb{R}$, then $\{f > 0\} = \mathbb{R}$).

$x > c \Rightarrow$ there exists $y \in D_a$ such that $c < y < x \Rightarrow f(x) \geq f(y) > a \Rightarrow x \in D_a$.

Hence $(c, \infty) \subseteq D_a$. Also it follows that $D_a \subseteq (c, \infty)$. Therefore $D_a = (c, \infty) \in \mathcal{B}_{\mathbb{R}}$. Thus we have

$$D_a = \begin{cases} [c, \infty) & \text{if } c = \inf D_a \in D_a \\ (c, \infty) & \text{if } c \notin D_a. \end{cases}$$

i.e. $D_a = \{f > a\}$ is an interval. Hence $\{f > a\} \in \mathcal{B}_{\mathbb{R}}$ for all $a \in \mathbb{R}$. From this we can see that $\{f \leq a\} \in \mathcal{B}_{\mathbb{R}}$ for all $a \in \mathbb{R}$. Therefore f is a Borel function.

¹ f is increasing if when ever $x < y$, then $f(x) \leq f(y)$

Lemma 0.1 (i) If f, g are Borel functions, so are $\alpha f, f + \alpha, f + g, fg, \max\{f, g\}, \min\{f, g\}$ where $\alpha \in \mathbb{R}$.

(ii) If $\{f_n\}$ is a sequence of Borel functions which is uniformly bounded, then $\sup_n f_n, \inf_n f_n$ are Borel functions.

(iii) If $\{f_n\}$ is a sequence of Borel functions such that

$$\lim_{n \rightarrow \infty} f_n(x) = f(x), x \in \mathbb{R}.$$

Then f is a Borel function

Proof is left as an exercise.

Example 0.3 (i) $f = I_B$, where B is a Borel set, then f is a Borel function. Call this as a Borel indicator function.

(ii) Any finite linear combinations of indicator functions are Borel functions. In particular step functions are Borel functions.

Now we have a result which gives us plenty of random variables.

Lemma 0.2 Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a Borel function and $X : \Omega \rightarrow \mathbb{R}$ is a random variable with respect to $\mathcal{F}$. Then $f \circ X$ is a random variable with respect to $\mathcal{F}$.

Proof. For $B \in \mathcal{B}_{\mathbb{R}}$,

$$(f \circ X)^{-1}(B) = X^{-1}(f^{-1}(B)) \in \mathcal{B}_{\mathbb{R}},$$

since $f^{-1}(B) \in \mathcal{B}_{\mathbb{R}}$. Hence $f \circ X$ is a random variable.

Example 0.4 The natural logarithm² $\ln : (0, \infty) \rightarrow \mathbb{R}$ is defined by

$$\ln x = \int_1^x \frac{1}{t} dt.$$

Note that $\ln$ is a strictly increasing and continuous function which is also a bijective map. We define the exponential function $e : \mathbb{R} \rightarrow (0, \infty)$ as the inverse of $\ln$. Then both are Borel measurable and hence $\ln X, e^X$ are random variables if X is a random variable.

²Natural logarithm of positive number is invented by John Napier in 1614. The definition we are using is the logarithmic property of the quadrature (determining area under a curve) rule for hyperbola. The notation log was introduced by Leibnitz in 1675 and subsequently in 1676, he invented the logarithmic property of the quadrature of hyperbola.

Chapter 4 : Conditional probability and independence

Key words: Conditional probability, law of total probability, Bayes theorem, independent events, independent σ -fields, independent random variables, lim sup and lim inf of events, Borel-Cantelli lemma.

In this chapter, we introduce the concepts of conditional probability and independence.

"Conditioning" is an important technique for computing probabilities. Conditioning is about writing down the probability of an event in terms of probability of the event given that some other events have already occurred. For using the above method, one need to understand these 'new' probabilities. Let us take an example to illustrate this. Suppose person ONE is inside a room and throw a die and secretly tell her friend TWO that an even number turned up. Now if the person THREE ask TWO how probable was the event $\{1, 2, 3\}$? TWO will rule out the outcomes 1 and 3 and scale it using even numbers to arrive at the new probability $\frac{1}{3}$ which is different from $\frac{1}{2}$. i.e., information changes the probabilities, these are called conditional probabilities. Formalizing the above 'slicing' and 'scaling' leads to the following definition of conditional probability.

Definition 4.1 Let $(\Omega, \mathcal{F}, P)$ be a probability space and $A \in \mathcal{F}$ be such that $P(A) > 0$. The conditional probability of an event $B \in \mathcal{F}$ given A denoted by $P(B|A)$ is defined as

$$P(B|A) = \frac{P(AB)}{P(A)}. \quad (0.2)$$

Define P_A on $\mathcal{F}$ as follows.

$$P_A(B) = P(B|A), \quad B \in \mathcal{F}.$$

Then $(\Omega, \mathcal{F}, P_A)$ is a probability space satisfying

$$P_A(A) = 1, \quad P_A(B) = 0 \text{ if } B \subseteq A^c.$$

(exercise)

Remark 0.1 *The above probability space may look useless but it is useful for processing the thought experiments of the following type. Note that $(\Omega, \mathcal{F}, P_A)$ corresponds to a random experiment which is obtained by adding*

the information that the event A has occurred. Hence by understanding directly this random experiment gives the probabilities $P_A(B)$ directly without using the formula (0.2) for $P_A(B)$. We will see an illustration of this in a moment.

Example 0.5 Let $\Omega = \{1, 2, 3, 4, 5, 6\}$ and $\mathcal{F} = \mathcal{P}(\Omega)$ and $P(\{i\}) = \frac{1}{6}$ for all i . Suppose, we got the information that 'even number' has occurred, then what are the new 'probabilities' i.e., the conditional probabilities of the events?

Set $A = \{2, 4, 6\}$. We need to compute P_A . Then from (0.2), we have

$$P_A(\{i\}) = \frac{1}{3}, \quad i = 2, 4, 6, \quad \text{and} \quad = 0, \quad i = 1, 3, 5.$$

Example 0.6 (Bridge hand) A pack of 52 cards are distributed among four players and is called a bridge hand. Find the probability of a balanced bridge hand of aces, i.e. each player gets an ace.

Define the following events. $A_{\clubsuit}$ denote the event that a player gets the ace of club, $A_{\clubsuit, \diamond}$, the event that two distinct players get an ace of club and diamond each, $A_{\clubsuit, \diamond, \heartsuit}$, the event that three distinct players get an ace of club, diamond and heart and $A_{\clubsuit, \diamond, \heartsuit, \spadesuit}$, the event that all players get an ace. Then

$$A_{\clubsuit, \diamond, \heartsuit, \spadesuit} \subset A_{\clubsuit, \diamond, \heartsuit} \subset A_{\clubsuit, \diamond} \subset A_{\clubsuit}.$$

and hence using $A_{\clubsuit, \diamond, \heartsuit, \spadesuit} = A_{\clubsuit, \diamond, \heartsuit, \spadesuit} \cap A_{\clubsuit, \diamond, \heartsuit} \cap A_{\clubsuit, \diamond} \cap A_{\clubsuit}$, we get (exercise)

$$P(A_{\clubsuit, \diamond, \heartsuit, \spadesuit}) = P(A_{\clubsuit, \diamond, \heartsuit, \spadesuit} | A_{\clubsuit, \diamond, \heartsuit}) P(A_{\clubsuit, \diamond, \heartsuit} | A_{\clubsuit, \diamond}) P(A_{\clubsuit, \diamond} | A_{\clubsuit}) P(A_{\clubsuit}).$$

(The above is a conditioning argument)

Now $P(A_{\clubsuit}) = 1$. To compute³ $P(A_{\clubsuit, \diamond} | A_{\clubsuit})$, observe that given $A_{\clubsuit}$, distributing diamond ace to one of the remaining three players is equivalent

³To compute the conditional probabilities, we will not use the definition of conditional probability instead we use an equivalent random experiment which gives raise to the probability space of conditional probabilities, as told in Remark 0.1. Note that given the information that the club ace is distributed to one player, we can think about the random experiment as an urn problem with 51 urns numbered 1 to 51 and a ball (identified with diamond ace) and our event is distributing the ball into one of the first 39 urns.

to placing the diamond ace in one of the 39 locations out of a total of 51 locations. Hence

$$\begin{aligned} P(A_{\clubsuit, \diamond} | A_{\spadesuit}) &= \frac{\text{no. of ways diamond ace can be placed in 39 locations}}{\text{no. of ways diamond ace can be placed in 51 locations}} \\ &= \frac{\binom{39}{1}}{\binom{51}{1}} = \frac{39}{51}. \end{aligned}$$

Similarly

$$P(A_{\clubsuit, \diamond, \heartsuit} | A_{\clubsuit, \diamond}) = \frac{26}{50}, \quad P(A_{\clubsuit, \diamond, \heartsuit, \spadesuit} | A_{\clubsuit, \diamond, \heartsuit}) = \frac{13}{49}.$$

Hence

$$P(A_{\clubsuit, \diamond, \heartsuit, \spadesuit}) = \frac{39 \times 26 \times 13}{51 \times 50 \times 49} = 0.11.$$

The above example illustrates that it is some times more easy to compute (or natural to specify) conditional probabilities and use them to specify the underlying probabilities, a reverse procedure!

Example 0.7 (*Probability in the game show "Let's make a deal"*) Here we look at a version of the game show "Let's make a deal" which made its debut on NBC Television network on December 30, 1963. Description of the game is the following. A prize is placed behind one of the three doors and are closed. Contestants of the show are aware of this. Contestant is asked to select a door (but is not going to open at the moment). Once the choice is made, the moderator of the show (Monty Hall) opens one of the remaining doors and display what is in it (He will only open a door which has no prize). Contestant now is given a chance to change the earlier choice. Question is, will the contestant stay with the earlier choice or not?

Without any loss of generality, assume that the contestant chose door no.1 (label chosen door as no.1).

Here take sample space as

$$\Omega = \{\diamond ab, \diamond ba, a \diamond b, b \diamond a, ab \diamond, ba \diamond\}.$$

(Here $\diamond ab$ denotes price $\diamond$ behind the first door and the 'worthless' a and b behind the doors 2 and 3 respectively. Other sample points are similraly interpreted.)

Question can be answered if we know the probability of ' $\diamond$ ' behind door 1' given the additional information of the object behind one of the doors 2 or 3.

Let us denote the event ' $\diamond$ ' behind door 1' by A . Let B_i denotes the event that Monty reveals door i . Also Monty Hall revealing object behind door 2 means behind door 2, object is either a or b . Then we can compute that $P(B_i|A) = \frac{1}{2}, i = 2, 3$ ⁴. Also $P(A) = \frac{1}{3}, P(B_2) = P(B_3) = \frac{1}{2}$. If Monty choose door 2, then using

$$P(A|B_2)P(B_2) = P(B_2|A)P(A) \quad (*)$$

we get $P(A|B_2) = \frac{1}{3}$. Similarly $P(A|B_3) = \frac{1}{3}$. Hence it is better to change the option, since you have 2/3rd chance of winning the prize by switching the door.

The formula (*) above written in the form

$$P(A|B_2) = \frac{1}{P(B_2)}P(B_2|A)P(A)$$

is called the Bayes' formula and is due to Bayes published posthumously in 1763 where he was addressing the Billiard table problem. A brief description of the Billiard table problem is as follows.

Marker : A white ball is thrown onto a rectangular billiard table of unit length. Its distance p from an edge (fixed) is assumed to be uniformly $(0, 1)$ distributed (prior). This position is marked with a line, and the white ball is removed.

Data : A number, say n , of red balls are thrown independently and at random onto the table.

Observation: We take note of the number of balls, say m landed to the left of the marker.

Goal: Compute the 'conditional probability distribution' for the true location of the marker p given the observed data.

Prior Distribution: Before any red balls are thrown, the probability density function for p is uniform, i.e., $f(p) = 1$ for $0 \leq p \leq 1$.

Likelihood Function: Given a specific position p , the probability of getting m balls on the left (each independently landing with probability p)

⁴If A occur, then door $i, i = 2, 3$ are worthless and hence Monty chooses either door with equal probability.

follows a binomial distribution:

$$P(\text{data} = m|p) = \binom{n}{m} p^m (1-p)^{n-m}.$$

Posterior Distribution: By applying Bayes' rule, the updated probability density of p after observing m and $n - m$ is proportional to the product of the prior and the likelihood:

$$f(p|\text{data}) \propto p^m (1-p)^{n-m} f(p).$$

The above example tells us how one can update the initial guess, (i.e., prior probabilities) using the likelihood of the data.

We have already seen 'finite' partitions of the sample space. A collection $\{A_1, A_2, \dots, A_N\}$ of events is said to be a countable *partition of Ω* , if

(i) A_n 's are pairwise disjoint

(ii) $\bigcup_{n=1}^N A_n = \Omega$.

Here N may be ∞ . If $N < \infty$, then partition is said to be finite partition and if $N = \infty$ ⁵, it is called a countably infinite partition.

Theorem 0.2 (*Law of total probability-discrete form*) Let $(\Omega, \mathcal{F}, P)$ be a probability space and $\{A_1, A_2, \dots, A_N\} \subseteq \mathcal{F}$ be a countable partition of Ω such that $P(A_n) > 0$ for all n . Then for $B \in \mathcal{F}$,

$$P(B) = \sum_{n=1}^N P(B|A_n) P(A_n).$$

Proof. Consider

$$\begin{aligned} \sum_{n=1}^N P(B|A_n) P(A_n) &= \sum_{n=1}^N \frac{P(B \cap A_n)}{P(A_n)} P(A_n) \\ &= \sum_{n=1}^N P(B \cap A_n) \\ &= P\left(B \cap \left(\bigcup_{n=1}^N A_n\right)\right) = P(B). \end{aligned}$$

The second last equality uses the countable additivity of probability when $N = \infty$ to get convergence of the series.

⁵When $N = \infty$, the collection is $\{A_1, A_2, \dots\}$

Theorem 0.3 (*Bayes'-Laplace Theorem*) Let $(\Omega, \mathcal{F}, P)$ be a probability space and $A, B \in \mathcal{F}$ such that $P(A), P(B) > 0$ and $P(A) < 1$. Then

$$P(A|B) = \frac{P(B|A)P(A)}{P(B|A)P(A) + P(B|A^c)P(A^c)}.$$

Proof. Apply Bayes' formula, we get

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}.$$

Now use law of total probability to complete the proof.

Example 0.8 (*Ballot problem*) In an election between two candidates, candidate I got n votes and II got 10 votes, where $n > 10$. Find the probability that the winner was leading throughout the ballot counting. The above Ballot problem is the simplest of the Ballot problems from Combinatorics.

For k, l non negative integers, let A_{kl}, F_{l+kk} respectively denote the events that I is leading throughout the counting and the last ballot was casted for I, when I has k votes and II has l votes. Also denote $P(A_{kl}) = p(k, l)$. Observe that

$$p(k, l) = 0, l \geq k, \quad p(k, 0) = 1 \text{ for all } k, l.$$

Then using law of total probability, we get

$$P(A_{kl}) = P(A_{kl}|F_{l+kk})P(F_{l+kk}) + P(A_{kl}|F_{l+kk}^c)P(F_{l+kk}^c).$$

Now

$$\begin{aligned} P(F_{l+k,k}) &= \frac{\text{no. of ways putting } k-1 \text{ votes in } l+k-1 \text{ locations}}{\text{no. of ways putting } k \text{ votes in } l+k \text{ locations}} \\ &= \frac{\binom{l+k-1}{k-1}}{\binom{l+k}{k}} = \frac{k}{l+k}. \end{aligned}$$

Hence

$$P(F_{l+k,k}^c) = \frac{l}{l+k}.$$

Now $P(A_{kl}|F_{l+k,k})$ can be viewed as the probability that I is leading from 1 to $l+k-1$ ballot counts when I has $k-1$ votes and II has k votes. Hence

$$P(A_{kl}|F_{l+k,k}) = p(k-1, l).$$

Similarly $P(A_{kl}|F_{l+k}^c)$ can be viewed as the probability that I leading from 1 to $l+k-1$ ballot counts where I has k votes and II has $l-1$ votes. Hence

$$P(A_{kl}|F_{l+k}^c) = p(k, l-1).$$

Hence we get the following recursive relation

$$P(k, l) = \frac{k}{l+k}p(k-1, l) + \frac{l}{l+k}p(k, l-1), k > l$$

with the boundary conditions $p(k, l) = 0, k \leq l$, $p(k, 0) = 1$ for all $k = 1, \dots, n$, $l = 0, 1, \dots, 10$. From the above, we have

$$\begin{aligned} p(1, 0) &= 1, p(1, 1) = 0, \\ p(2, 0) &= 1, p(2, 1) = \frac{1}{3}, p(2, 2) = 0, \\ p(3, 0) &= 1, p(3, 1) = \frac{1}{2}, p(3, 2) = \frac{1}{5} \\ p(4, 0) &= 1, p(4, 1) = \frac{3}{5}, p(4, 2) = \frac{1}{3}, p(4, 3) = \frac{1}{7}, p(4, 4) = 0. \end{aligned}$$

Hence, we guess that

$$p(k, l) = \frac{k-l}{k+l}, k \geq l, l+k \geq 2.$$

Now assume that the above is true for $l+k = 0, 1, 2, \dots, m$, then for (k, l) with $l+k = m+1$, it immediately follows from the recursive relation that

$$p(k, l) = \frac{k-l}{k+l}, k \geq l, l+k = m+1.$$

This implies by induction that the required probability is $\frac{n-10}{n+10}$.